

Carbon-nitrogen complex as an alternative to the T center in SiJ. K. Nangoi^{1,*}, M. E. Turiansky^{1,2}, and C. G. Van de Walle¹¹*Materials Department, University of California, Santa Barbara, California 93106, USA*²*U.S. Naval Research Laboratory, Washington, DC 20375, USA*

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We present a first-principles study of a carbon-nitrogen (CN) impurity complex in silicon as an isoelectronic alternative to the T center $[(\text{CCH})_{\text{Si}}]$. The latter has been pursued for applications in quantum information science, yet its sensitivity to the presence of hydrogen is still problematic. Our proposed complex has no hydrogen, thereby eliminating this issue. First, we show that the CN complex is stable against decomposition into substitutional and interstitial defects. Next, we show that due to being isoelectronic to the T center, the CN complex has a similar electronic structure, and therefore could be used in similar applications. We assess several low-energy configurations of the CN complex, finding $(\text{CN})_{\text{Si}}$ to be stable and have the largest Debye-Waller factor. We predict a zero-phonon line (ZPL) of 828 meV (in the telecom S band) and a radiative lifetime of 4.2 μs , comparable to the T center. Due to the presence of a bound exciton, choice of the exchange-correlation functional and also supercell-size scaling of the ZPL and transition dipole moment require special scrutiny; we rigorously justify our extrapolation schemes that allow computing values in the dilute limit.

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Point defects in semiconductors or insulators are being studied for quantum information science applications, including as spin qubits for quantum computing [1–3] and single-photon emitters for quantum networks [2–4]. For the former, the spin coherence time is the main metric, as it determines how long quantum information remains coherent. For networking, single photons carry the quantum information [5,6]. A high Debye-Waller (DW) factor is desired so that most photons are emitted into the zero-phonon line (ZPL), for which the photons are in a well-defined quantum state [6].

The nitrogen-vacancy (NV) center in diamond has been studied extensively for these applications. It has a triplet ground state with spin coherence time exceeding milliseconds [7], but a DW factor of only $\sim 3\%$ [8]. Alternatives to the NV center have been studied, such as the silicon-vacancy center in diamond [9], carbon vacancy and silicon vacancy in cubic boron nitride [10], or color centers in silicon [3,11,12]. Silicon is an attractive host material because it offers the prospect of integration with Si-based electronics [11]; it is also far easier to grow and process than diamond [7]. The T center in Si, a complex in which two carbons and one hydrogen substitute on a Si site [denoted by $(\text{CCH})_{\text{Si}}$, Fig. 1(a)], has been intensively pursued experimentally due to its high spin coherence times, large DW factor, and emission in the O band of telecom wavelengths [11,13–17].

Despite its promise, formation and stability of the T center remain a concern; it was found to be “prone to (de)hydrogenation and so requires very precise annealing conditions (temperature and atmosphere) to be efficiently formed” [12]. Identifying alternatives to the T center, preferably without H, would thus be beneficial.

In this Letter, we propose a carbon-nitrogen complex as an analog of the T center. Replacing the hydrogen and one carbon atom in $(\text{CCH})_{\text{Si}}$ with a nitrogen atom keeps the center isoelectronic with the T center, since N contains the same number of electrons and protons as $\text{C} + \text{H}$. The absence of H makes this complex more stable with regard to (de)hydrogenation. Like H, N has nonzero nuclear spin and can thus be exploited to store quantum information. We demonstrate the stability of this center by calculating the formation and decomposition energies, and thoroughly assess its electronic structure and optical properties, including the Huang-Rhys/DW factors, energy of the ZPL transition, and radiative lifetime. Our results show that the CN center is a promising candidate for applications in quantum information science.

Our first-principles studies are based on density-functional theory (DFT) with projector-augmented wave potentials [18,19] as implemented in the Vienna *Ab initio* Simulation Package (VASP) [20,21], with a plane-wave cutoff of 400 eV. We use the hybrid functional of Heyd, Scuseria, and Ernzerhof (HSE) [22,23] with the default mixing parameter of 25%; for select results (as described below) we also employ the PBE0 (Perdew, Burke, and Ernzerhof) hybrid functional [24–26] with a mixing parameter of 13.6%. For the primitive cell with a Brillouin-zone sampling mesh of $11 \times 11 \times 11$, we find an HSE lattice constant of 5.433 Å and a band gap of 1.15 eV, both in agreement with experiment (5.431 Å [27] and 1.17 eV at 0 K [28]). We model the defect in a supercell geometry, using the Γ point to sample the Brillouin zone. Most of our results are presented for a 512-atom supercell ($4 \times 4 \times 4$ multiple of the conventional cubic cell), allowing us to make direct comparison with previous work [12]. We also use up to 1000-atom supercells ($5 \times 5 \times 5$) to better describe the bound excitons that are present in the excited states of the centers

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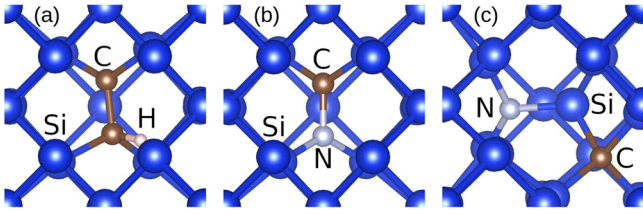


FIG. 1. Structure of (a) T center, $(\text{CCH})_{\text{Si}}$, (b) $(\text{CN})_{\text{Si}}$, and (c) $\text{CSi}(\text{NSi})_{\text{Si}}$.

studied here. Structural optimizations are performed until the forces are less than $0.01 \text{ eV/\AA}$.

Previous first-principles calculations identified two possible configurations of the CN complex: (1) a C-N split interstitial $(\text{CN})_{\text{Si}}$ [29] [Fig. 1(b)], which can be thought of as replacing C-H with N in the T center $(\text{CCH})_{\text{Si}}$, and (2) a complex of a substitutional C atom and a N-Si split interstitial, $\text{CSi}(\text{NSi})_{\text{Si}}$ [30,31] [Fig. 1(c)].

Figure 2 shows our calculated formation energies E^f as a function of the Fermi level for both CN structures, and also for the T center. E^f is given by [32]

$$E^f[X^q] = E_{\text{tot}}[X^q] - E_{\text{tot}}[\text{Si}] - \sum_i n_i \mu_i + qE_F + E_{\text{corr}},$$

where $E_{\text{tot}}[X^q]$ is the total energy of the supercell containing the defect X in charge state q , $E_{\text{tot}}[\text{Si}]$ is the total energy of the equivalent supercell containing perfect host material, n_i is the number of atoms of type i added to ($n_i > 0$) or removed from ($n_i < 0$) the supercell, μ_i is the chemical potential of atom type i , E_F is the Fermi level, and E_{corr} is a finite-size correction for charged defects [33]. For the μ_i we use the total energies per atom of bulk Si, diamond, H_2 , and N_2 .

Our results for the T center agree with Ref. [12]. The $+1$ charge state of the T center and $(\text{CN})_{\text{Si}}$ are not stable in the gap, as seen in Fig. 2. We find that the -1 charge state of the T center, $(\text{CN})_{\text{Si}}$, and $\text{CSi}(\text{NSi})_{\text{Si}}$, and also the $+1$ charge state of $\text{CSi}(\text{NSi})_{\text{Si}}$, all have net-zero spin (singlet) (see Fig. S1 in the Supplemental Material (SM) [34]) (see also Refs. [35–44] therein), precluding their use as spin qubits. We also note that

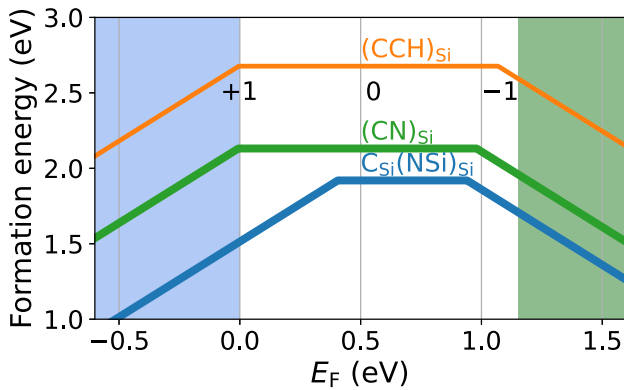


FIG. 2. Defect formation energies as a function of Fermi level for the T center (orange), $(\text{CN})_{\text{Si}}$ (green), and $\text{CSi}(\text{NSi})_{\text{Si}}$ (blue). Blue and green shades indicate valence- and conduction bands, respectively. The valence-band maximum (VBM) is set to 0, and the conduction-band minimum (CBM) is at 1.15 eV , our calculated Si band gap.

TABLE I. Decomposition energies ΔE^f for $(\text{CN})_{\text{Si}}^0$ and $[\text{CSi}(\text{NSi})_{\text{Si}}]^0$.

	ΔE^f (eV)	
	$(\text{CN})_{\text{Si}}^0$	$[\text{CSi}(\text{NSi})_{\text{Si}}]^0$
$\rightarrow \text{C}_{\text{Si}}^0 + (\text{NSi})_{\text{Si}}^0$	0.61	0.82
$\rightarrow \text{C}_{\text{Si}}^{-1} + (\text{NSi})_{\text{Si}}^{+1}$	1.33	1.54
$\rightarrow \text{C}_{\text{Si}}^{+1} + (\text{NSi})_{\text{Si}}^{-1}$	1.69	1.90
$\rightarrow (\text{CSi})_{\text{Si}}^0 + \text{N}_{\text{Si}}^0$	3.12	3.33
$\rightarrow (\text{CSi})_{\text{Si}}^{-1} + \text{N}_{\text{Si}}^{+1}$	3.65	3.86
$\rightarrow (\text{CSi})_{\text{Si}}^{+1} + \text{N}_{\text{Si}}^{-1}$	3.70	3.91

the excited state of, e.g., the -1 charge state involves exciting an electron to the conduction-band minimum (CBM), leaving a neutral defect center behind. The electron will thus feel no Coulomb attraction and hence cannot act as a single-photon emitter. Similar arguments apply to the $+1$ charge state. These charge states are therefore less useful for quantum information applications. We thus focus on the neutral charge state, which is stable over the majority of the band gap for $(\text{CN})_{\text{Si}}$, with the negative charge state occurring only for Fermi levels within 0.17 eV of the CBM. $[\text{CSi}(\text{NSi})_{\text{Si}}]^0$ is stable for Fermi levels between 0.41 and 0.94 eV .

The stability of alternate atomic configurations in the case of the CN center inspired us to investigate whether the CCH center could also be stable in other structures; we found these to be 0.6 – 2.5 eV higher in energy than the accepted $(\text{CCH})_{\text{Si}}$ structure of the T center (see Sec. S2 of the SM [34]).

We assess the stability of the CN defects [both $\text{CSi}(\text{NSi})_{\text{Si}}$ and $(\text{CN})_{\text{Si}}$] with respect to decomposition into constituent defects (substitutions and interstitials) by calculating the decomposition energy $\Delta E^f \equiv \sum E^f[\text{products}] - E^f[\text{CN defect}]$; a positive energy indicates the reaction is endothermic. For C and N interstitials, the split-interstitial configurations are found to be lowest in energy, consistent with Refs. [29,45–48] (see Sec. S2 in the SM [34] for atomic structures and formation energies).

Table I shows our results for decomposition energies, taking into account that the overall charge state should remain neutral. All decomposition energies are positive, meaning both configurations are stable against all considered decompositions. The ΔE^f for $(\text{CN})_{\text{Si}}^0$ are $\sim 0.2 \text{ eV}$ lower than those for $[\text{CSi}(\text{NSi})_{\text{Si}}]^0$, because the formation energy of $(\text{CN})_{\text{Si}}^0$ is slightly higher than that of $[\text{CSi}(\text{NSi})_{\text{Si}}]^0$ (Fig. 2). Decomposition into C_{Si}^0 and $(\text{NSi})_{\text{Si}}^0$ has the smallest energy, with a value that is comparable to the lowest decomposition energy (0.80 eV) for $(\text{CCH})_{\text{Si}}$ calculated in Ref. [12]. Furthermore, as mentioned above, the absence of hydrogen in the CN defect is advantageous.

We use the climbing image nudged elastic band method [49] to calculate the migration barrier of $(\text{NSi})_{\text{Si}}^0$ (the most mobile constituent), resulting in 0.68 eV (see Sec. S3 in the SM [34]). We can thus estimate the barrier height of the lowest-energy decomposition reaction to be 1.50 eV for $[\text{CSi}(\text{NSi})_{\text{Si}}]^0$ and 1.29 eV for $(\text{CN})_{\text{Si}}^0$.

We now study the electronic structure by analyzing the spin-polarized Kohn-Sham states and wave functions in both

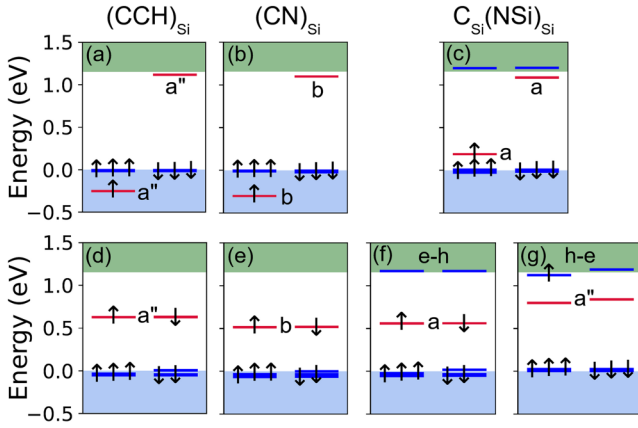


FIG. 3. Ground- (top row) and excited-state (bottom row) Kohn-Sham states for the neutral charge state of the T center [(a), (d)], (CN)_{Si} [(b), (e)], and C_{Si}(NSi)_{Si} [(c), (f), (g)]. “e-h” (“h-e”) labels the localized-electron (hole) case. Blue and green shades indicate valence and conduction bands. Red levels are the states associated with the defect, and blue levels are valence- and conduction-band states.

the ground- and excited electronic states. Figure 3 compares (CN)_{Si}⁰ and [C_{Si}(NSi)_{Si}]⁰ with (CCH)_{Si}⁰ (the T center, for which our results agree with Ref. [12]). In the ground state of the T center, the a'' antibonding state associated with the C_{1h} symmetry undergoes exchange splitting: the occupied spin-up state is below the VBM and the unoccupied spin-down state lies just below the CBM. The ground states of the CN centers are qualitatively similar, with exchange splitting in the b antibonding state for (CN)_{Si} (C_{2v} symmetry) and in the a state for C_{Si}(NSi)_{Si} (C_1 symmetry), although the occupied spin-up a state is now above the VBM. The unoccupied states for all three defects [Figs. 4(a)–4(c)] and the occupied state for C_{Si}(NSi)_{Si} [Fig. 4(d)] are localized at the defect site.

To model excited states, we use the constrained occupation Δ SCF (self-consistent field) approach [50], where we excite a spin-down electron from the VBM to the unoccupied defect state, constrain the electron occupation, and reoptimize the structure. The resulting structure has an in-gap state with both spin channels occupied [Figs. 3(d)–3(f)] that is localized at the defect site [Figs. 4(e)–4(g)] (maintaining the symmetry of the ground state), and leaves a hole with hydrogenic nature bound to the localized electron. The localized electron and the hole form a bound exciton, where, within hydrogenic effective-mass theory, the hole approximately has an effective Bohr radius of ~ 13 Å as shown in Sec. S4 of the SM [34].

For C_{Si}(NSi)_{Si}, it is also possible to excite a spin-up electron from the in-gap defect state into the CBM, resulting in an exciton with a localized hole and a hydrogenic electron [Figs. 3(g) and 4(h)]. The corresponding effective Bohr radius for the electron is approximately ~ 6 Å (see Sec. S4 of the SM [34]). In contrast, for both the T center and (CN)_{Si}, exciting the electron from the a'' or b state below the VBM into the CBM results in an excited state where the hole is at the VBM instead of being localized at an in-gap defect state.

We note that the localized-hole excited state of C_{Si}(NSi)_{Si} has C_{1h} symmetry, unlike the C_1 symmetry for both the ground state and the localized-electron excited state, as apparent in

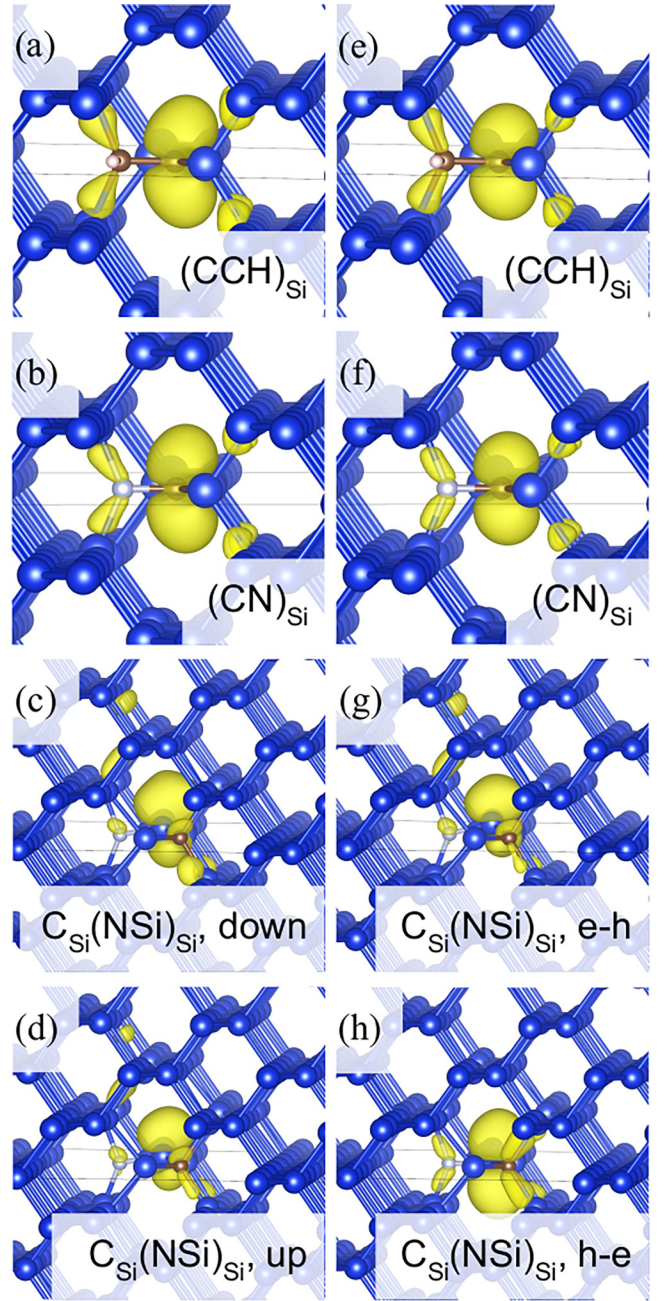


FIG. 4. Isosurfaces (yellow) of real-space Kohn-Sham probability densities for the neutral charge state of the ground-state in-gap empty spin-down state [(a)–(c)] and filled spin-up state [(d)], and of the excited-state in-gap filled spin-down state [(e)–(g)] for localized-electron case and empty spin-up state [(h)] for localized-hole case. Blue circles are Si; brown, C; light blue, N; and pink, H.

the charge density [Fig. 4(h)]. This is because, as seen in the figure, the C, N, and Si bonded to both C and N now form a mirror plane, unlike in the ground state [Figs. 4(c) and 4(d)] and the localized-electron excited state [Fig. 4(g)].

Table II shows our calculated Huang-Rhys factors $S = E_r/(\hbar\Omega)$ and DW factors $\exp(-S)$; E_r and Ω , respectively, are the relaxation energy and phonon frequency in the electronic ground state within the one-dimensional approximation [51–53]. Details are in Sec. S5 of the SM [34]. Our

TABLE II. Relaxation energies (E_r), ground-state phonon frequencies (Ω), and Huang-Rhys (S) and Debye-Waller (DW) factors.

	E_r (eV)	$\hbar\Omega$ (meV)	S	DW (%)
(CCH) _{Si}	0.079	33	2.4	9.2
(CN) _{Si}	0.107	35	3.0	5.0
C _{Si} (NSi) _{Si} e-h	0.157	28	5.6	0.4
C _{Si} (NSi) _{Si} h-e	0.297	28	10.5	0.003

calculated DW factor for the T center is $\sim 9\%$; Ref. [54] reports a calculated value of 16.5%, while the value obtained from photoluminescence data is 23% [11]. The discrepancy with Ref. [54] could be due to differences in E_r or Ω . The discrepancy with experiment could be because our DW factors are based on a one-dimensional model, which underestimates the more realistic, multidimensional model [55], so we focus here on *trends*. The DW factors for C_{Si}(NSi)_{Si} are below 1%, limiting their usefulness for single-photon emitters. In the following, we focus on (CN)_{Si}.

The energy of the ZPL transition E_{ZPL} is calculated as the total-energy difference between the excited state and the ground state. Using a 512-atom supercell, we find $E_{ZPL} = 981$ meV for the T center, in agreement with Ref. [12]. However, as seen in Fig. 5 the calculated E_{ZPL} depends on the supercell size because our supercells are not large enough to completely fit the hydrogenic wave functions (Fig. S5 in the SM [34]) [56]. The dependence of E_{ZPL} on supercell size is well described by a linear fit to the inverse of the supercell volume (Fig. 5). An extrapolation to the dilute limit based on our HSE values produces a value of 1064 meV, overestimating the experimentally measured ZPL of 935 meV [11].

An investigation of the cause of this deviation indicated that the ZPL values are sensitive to the DFT functional. We

found that PBE0 gives a better extrapolated ZPL for the T center, and therefore chose it to improve predictions for the CN defects.

A PBE0 mixing parameter of 13.6% yields a lattice constant of 5.446 Å and a band gap of 1.23 eV, which reproduces the experimental $T = 0$ gap of 1.17 eV [28] plus the zero-point renormalization of 60 meV [57]. We found that PBE0 geometries are very close to those obtained with HSE, and hence we used single-shot PBE0 (i.e., only performing electronic structure optimization) using the relaxed HSE geometries (scaled according to the slight difference in lattice constant). The accuracy of this approach is demonstrated in Table S1 of the SM [34].

As shown in Fig. 5, extrapolation of the PBE0 values to the dilute limit yields 918 meV for the T center [(CCH)_{Si}], significantly closer to experiment (935 meV, Ref. [11]) than the HSE value of 1064 meV.

For (CN)_{Si}, we find an extrapolated PBE0 value of 828 meV (in the S band of the telecom wavelength region [58]); see Sec. S7 for the values for C_{Si}(NSi)_{Si}. A number of luminescence lines with comparable energies have been observed in Si, at 829.8 meV [59], 844 meV [60], and 856 meV; the latter was suggested to be associated with interstitial C [59]. Experimental ZPLs ranging from 746 to 772 meV have been attributed to complexes containing C and N [59,61–65]; however, no reliable microscopic identification of the structure of these complexes is available, and the cited references indicate that some may involve additional impurities such as oxygen.

Finally, we calculate the radiative lifetime τ of the ZPL transition using the Weisskopf-Wigner formula [6,66,67]:

$$\frac{1}{\tau} = \left(\frac{\mathcal{E}_{\text{eff}}}{\mathcal{E}_0} \right)^2 \frac{n_r}{3\pi\epsilon_0 c^3 \hbar^4} (E_{ZPL})^3 |\mu|^2, \quad (1)$$

where n_r is the refractive index of the host material (3.38 for Si [27]), c is the speed of light, and μ is the transition dipole moment (TDM). The prefactor $(\mathcal{E}_{\text{eff}}/\mathcal{E}_0)^2$ accounts for local-field effects [6,66], which tend to increase the rate. Here we take $\mathcal{E}_{\text{eff}} \approx \mathcal{E}_0$, which is a common approximation [6,68].

Calculating μ is challenging because it requires accurately describing the hydrogenic wave function corresponding to the bound exciton. We therefore do not attempt to calculate μ in the excited state, but we approximate μ by scaling the TDM calculated in the *ground* state, μ^0 , to account for the hydrogenic nature:

$$|\mu|^2 \approx t |\mu^0|^2. \quad (2)$$

Here $t = \tilde{V}/[\pi(a_0^*)^3]$, where $\tilde{V}$ is the supercell volume and a_0^* is the effective Bohr radius of the hydrogenic wave function. For details, see Sec. S8 of the SM [34] [69].

Using the approximate μ and the extrapolated ZPLs, we find very similar radiative lifetimes of 4.70 μs (which agrees well with 4.9 μs deduced from experiments in Ref. [70]) and 4.18 μs for the T center and (CN)_{Si}. [See Table S2 for the values for C_{Si}(NSi)_{Si}.]

In conclusion, we propose the CN complex as a hydrogen-free alternative to the T center for similar quantum applications. We have shown that both (CN)_{Si} and

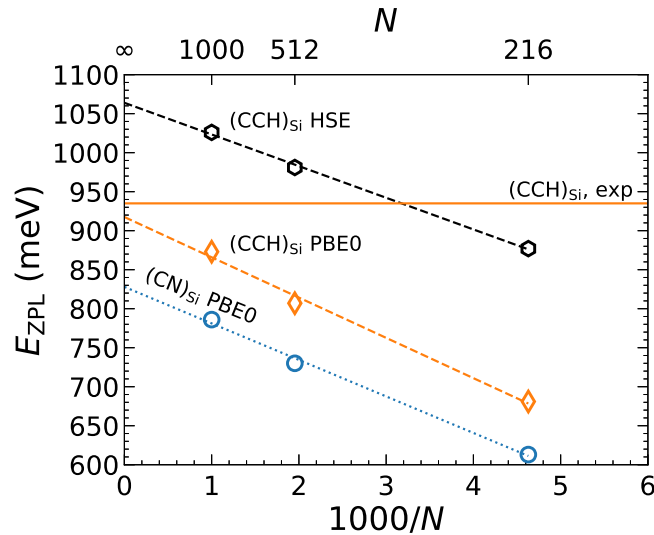


FIG. 5. Zero-phonon line energies for the T and CN centers, calculated using either single-shot PBE0 or HSE and plotted as a function of inverse supercell size (N is the number of atoms in the defect-free supercell). The lines are linear fits. The horizontal orange line represents the measured ZPL of the T center, 935 meV from Ref. [11].

$C_{Si}(NSi)_i$ are stable against decomposition to C and N substitutional/interstitial defects, and have electronic structures similar to the T center: a neutral charge state that is stable in the band gap, similar Kohn-Sham eigenvalues and eigenstates in both the ground- and excited states, and an excited state consisting of a bound exciton with a localized electron and a hydrogenic hole. Additionally, $C_{Si}(NSi)_i$ has an excited state consisting of a bound exciton with a localized *hole* and a hydrogenic *electron*. We carefully handle the supercell-size scaling for the ZPL and propose an extrapolation procedure for the radiative lifetimes, allowing us to obtain results in the dilute limit. We find the lifetime for $(CN)_{Si}$ to be similar to the $(CCH)_{Si}$ center. These results, combined with the fact that the predicted ZPL of $(CN)_{Si}$ is in the telecom S band, render the $(CN)_{Si}$ center a promising hydrogen-free alternative to the T center.

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The data that support the findings of this article are openly available [72].

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